



ESG disclosure and corporate cost stickiness: Evidence from supply-chain relationships

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ABSTRACT

We investigate the effect of Environmental, social, and governance (ESG) disclosures by customer firms on suppliers' cost stickiness in a supply-chain setting. Our findings indicate that ESG disclosures can mitigate suppliers' cost stickiness, mainly due to reduced optimistic expectations from suppliers' management. Further analysis reveals that the negative relationship intensifies with increased geographical and social distance between suppliers and customers, and when they belong to different industries.

1. Introduction

In the current economic context, sustainable development is recognized as an essential investment principle, compelling firms to focus not only on financial but also social performance. Consequently, stakeholders consider ESG (Environmental, Social, Governance) risks and opportunities in their decision-making. Despite abundant research on ESG disclosure's influence on various stakeholders, its direct impact on supply chain dynamics, particularly suppliers, is often overlooked. In this study, we focus on the stakeholders in supply-chain relationships and aim to investigate suppliers' responses to customers' ESG disclosure. Specifically, we examine whether and how ESG disclosures by customer firms affect suppliers' cost stickiness.

Several characteristics of the supply-chain setting make it particularly suitable for studying the impact of ESG disclosure on stakeholders. First, relationships between suppliers and customers have significant economic influence on business activities (Carvalho et al., 2020). Second, suppliers have an incentive to pay attention to their customers' ESG information to make sure that they meet the requirement of their ESG practices (Dai et al., 2021).

Using data from Chinese listed companies, our study reveals that ESG disclosures by customer firms can decrease suppliers' cost stickiness. Our analysis points to reduced optimistic expectations from suppliers' management as the primary mechanism. Additionally, the effect is more pronounced under conditions of greater geographical and social

distance between suppliers and customers, as well as across different industries. This paper makes several contributions. First, our findings add to the literature on the economic consequences of ESG disclosure. Whereas the existing literature focuses on how ESG disclosure affects investors, creditors, and other stakeholders (El Ghoul and Karoui, 2017; Christensen et al., 2022), our results prove that ESG disclosures by customer firms can affect suppliers' cost decisions.

Second, this paper provides new evidence for information spillover effects in supply-chain relationships. Previous studies explore the information spillover effects of customer operating performance information, such as bankruptcy announcements, surplus announcements, and releases of short-selling information (Cho et al., 2020; Chen et al., 2022; Chiu et al., 2022). This paper, on the other hand, provides evidence that ESG disclosures by customer firms generate information spillover effects in supply-chain relationships.

Finally, this study enriches the literature on supply-chain ESG. Whereas several studies have pointed out that ESG behavior by customer firms is positively correlated with that of their suppliers (Schiller, 2018; Dai et al., 2021), this paper focuses on ESG information spillovers in supply-chain relationships.

2. Hypothesis development

The prevailing view is that management's optimistic expectations, agency problems, and adjustment costs are the three main influences on

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cost stickiness (Dierynck et al., 2012; Banker and Byzalov, 2014). This paper identifies lowered optimistic expectations on the part of the management of suppliers as the underlying mechanism through which ESG disclosures by customer firms affects suppliers' cost stickiness.

ESG disclosures contain valuable information about firms' business fundamentals. Therefore, the more ESG information about customer firms is available, the easier it is for suppliers to develop an objective understanding of their customers' business situations and prospects. This can help suppliers correct optimistic forecasting biases resulting from management overestimation of future demand. When customer demand declines, the suppliers' management will not blindly and confidently treat the decline as a normal fluctuation but will instead cut idle resources based on actual demand conditions, which may help to reduce suppliers' cost stickiness.

The discourse on supply-chain management frequently addresses the role of blockchain technology in enhancing disclosure practices (Dutta et al., 2020; Abidi et al., 2021; Zheng et al., 2022), drawing parallels with the impacts of ESG disclosure on supply-chain transparency and reduction of information asymmetry. While blockchain emphasizes technological solutions for secure and efficient information sharing, ESG disclosures offer additional, valuable insights on a firm's environmental, social, and governance aspects. This distinction highlights blockchain's focus on improving information's traceability and security, and ESG disclosure's capacity to inform suppliers about potential risks and opportunities related to their customers, contributing to a more accurate risk assessment and moderating management's optimistic expectations.

Based on this discussion, we propose our hypothesis as follows:

H1: ESG disclosures by customer firms is negatively associated with suppliers' cost stickiness.

3. Research methods

According to Banker et al., 2014, Holzhacker et al. (2015) and Dai et al. (2021), we use the following model to test the hypothesis H1:

$$\Delta \ln SGA = \beta_0 + \beta_1 \times \Delta \ln Rev + \beta_2 \times D \times \Delta \ln Rev + \beta_3 \times ESG \times D \times \Delta \ln Rev + \beta_j \times Controls \times D \times \Delta \ln Rev + \sum Industry + \sum year + \varepsilon, \quad (1)$$

where $\Delta \ln SGA$ is the dependent variable, measuring the change in the natural logarithm of SG&A (Selling, general and administrative) fee between year $t-1$ and t , $\Delta \ln Rev$ is the change in the natural logarithm of net sales revenue between year $t-1$ and t , D is the value of one when the sales revenue in year t is lower than that in year $t-1$ and 0 otherwise. ESG is the customers' ESG disclosure score from Bloomberg. $Controls$ are the control variables. $\sum Industry$ represents the industry fixed effect, and $\sum year$ represents the year fixed effect. Appendix A shows the sample selection process. Appendix B presents the detailed definitions. Appendix C presents the descriptive statistics for all variables in the regression analysis.

In model (1), β_2 is the proxy for cost stickiness. If the sample exhibits cost stickiness, β_2 will be significantly negative. If H1 is valid, the coefficient β_3 will be significantly positive.

4. Results

4.1. Regression analysis

Table 1 reports the results of the regression. We include a simplified model without control variables in Columns (1) and (2) of Table 1. As shown in Columns (1)–(2) of Table 1, the coefficients of $D \times \Delta \ln Rev$ are significantly negative at 1 % and 10 % levels respectively, proving the presence of cost stickiness in sample firms. The coefficients of $ESG \times D \times \Delta \ln Rev$ are significantly positive at 1 % and 5 % levels respectively, thus supporting H1, indicating that ESG disclosures by customer firms can reduce suppliers' cost stickiness.

Table 1
Regression results.

Dependent variable: $\Delta \ln SGA$	(1)	(2)
$\Delta \ln Rev$	0.620*** (5.00)	0.652*** (5.32)
$D \times \Delta \ln Rev$	−1.643*** (−4.48)	−3.400* (−1.95)
$ESG \times D \times \Delta \ln Rev$	0.030*** (3.19)	0.022** (2.34)
$AI \times D \times \Delta \ln Rev$		−0.070 (−1.18)
$EI \times D \times \Delta \ln Rev$		0.071 (1.60)
$AFCFF \times D \times \Delta \ln Rev$		0.788*** (3.38)
$Sucdec \times D \times \Delta \ln Rev$		0.052 (0.45)
$GDPgrow \times D \times \Delta \ln Rev$		6.291*** (3.08)
$Size \times D \times \Delta \ln Rev$		−0.014 (−0.27)
$Lev \times D \times \Delta \ln Rev$		0.202 (0.69)
$ROA \times D \times \Delta \ln Rev$		0.117 (0.33)
$SOE \times D \times \Delta \ln Rev$		−0.300 (−1.56)
$Size_c \times D \times \Delta \ln Rev$		0.076 (1.19)
$Lev_c \times D \times \Delta \ln Rev$		−0.735 (−1.11)
$ROA_c \times D \times \Delta \ln Rev$		−2.217* (−1.77)
$SOE_c \times D \times \Delta \ln Rev$		0.127 (0.73)
Constant	0.065 (0.47)	0.097 (0.70)
Year	Yes	Yes
Industry	Yes	Yes
Observations	1399	1399
Adjusted R ²	0.357	0.391

Notes: This table reports the estimation results of model (1). Column (1) shows the regression results without the control variables. Column (2) shows the regression results with the control variables. Below the coefficient estimates, in parentheses, are t-values. *, **, and *** indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

4.2. Robustness tests

4.2.1. Endogenous mitigation

Heckman selection model

We use the Heckman selection model to address the self-selection problem (Heckman, 1979). In the first-stage model, we add the mean value of ESG of other firms in the same year and region as the additional control variable and run a Probit regression to predict the likelihood of a firm disclosing customer information. Then, we recover the inverse mills ratio (IMR). In the second stage, we augment model (1) with the IMR to control for self-selection bias. We report the results in Column (1) of Table 2. The coefficient of $ESG \times D \times \Delta \ln Rev$ continues to be negatively significant at the 5 % level.

Instrumental variable estimation

Although the baseline regression model includes a series of control variables, it is often unable to account for all variables that can impact suppliers, potentially leading to endogeneity issues caused by omitted variables. We address this problem by using the mean of customers' ESG disclosure in the same year and province (ESG_{IV}) as the instrument variable (Di et al., 2020). The results of the instrumental variable method are reported in Columns (2) and (3) of Table 2. Column (2) presents the regression results of the first stage. Column (3) presents the regression results of the second stage. The results remain robust to using the instrumental variable.

Table 2
Endogenous mitigation and control the effect of blockchain technology.

	Endogenous solution			Add control variable
Dependent variable: $\Delta \ln \text{SGA}$ ($\text{ESG} \times D \times \Delta \ln \text{Rev}$)	Heckman selection model	Instrumental variable		Add BLK as control variable
		First stage	Two stage	
	(1)	(2)	(3)	(4)
$\Delta \ln \text{Rev}$	0.726*** (6.16)	−0.116 (−0.67)	0.708*** (4.29)	0.651*** (7.36)
$D \times \Delta \ln \text{Rev}$	−3.925** (−2.33)	−65.293*** (−4.59)	−1.849 (−0.68)	−3.386** (−2.14)
$\text{ESG} \times D \times \Delta \ln \text{Rev}$	0.026** (2.53)		0.048** (2.43)	0.022*** (2.74)
$\text{BLK} \times D \times \Delta \ln \text{Rev}$				0.399 (0.83)
$\text{ESG}_{IV} \times D \times \Delta \ln \text{Rev}$		0.800*** (7.60)		
IMR	0.031 (0.76)			
Constant	0.067 (0.51)	0.029 (0.35)	0.074* (1.78)	
Underidentification test		Chi-sq(1)=35.20 (p-value=0.000)		
Weak identification test		F statistic=74.51 (10 % maximal IV size=16.38)		
Controls	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
Observations	1399	1399	1399	1399
Adjusted R ²	0.461	0.972	0.404	0.438

Notes: This table reports the results of the endogenous solution and controlling the effect of blockchain technology. Column (1) shows the regression results using the Heckman two-stage method. Column (2) and (3) show the regression results using instrument variable. Column (4) shows the regression results of controlling for the variable of blockchain technology development (BLK). Below the coefficient estimates, in parentheses, are t-values. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

4.2.2. Control the effect of blockchain technology

The varying levels of blockchain technology development may affect the efficiency of information transmission across supply-chain. To eliminate the interference of this factor, we have controlled for the variable of blockchain technology development ($\text{BLK}^{1 \text{ and } 2}$) in our main regression. The regression results, as shown in column (4) of Table 2, indicate that the coefficient of $\text{ESG} \times D \times \Delta \ln \text{Rev}$ continues to be negatively significant at the 1 % level.

4.2.3. Placebo test

Another potential endogeneity issue in this study is that the effect of ESG disclosures by customer firms on suppliers' cost stickiness may not due to the establishment of supply-chain relationships between the two parties, but result from other unobservable random omitted variables. We use placebo test to mitigate this issue. Specifically, we conduct the regression by randomly matching customers to suppliers (RCUS), randomly matching suppliers to customers (RSUP), and simultaneously randomly matching both customers and suppliers (RCUSSUP). The results in Table 3 show that the significantly effect of ESG disclosures by customer firms on suppliers' cost stickiness is rare and robust rather than a random regression's result.

4.3. Mechanism analysis

The main factors influencing cost stickiness are management's optimistic expectations, agency problems, and adjustment costs.

¹ If blockchain technology is mentioned in the annual reports of companies, then BLK is assigned a value of 1; if not mentioned, it is assigned a value of 0.

Table 3
Placebo test.

Dependent variable: $\Delta \ln \text{SGA}$	(1) RCUS	(2) RSUP	(3) RCUSSUP
$\Delta \ln \text{Rev}$	0.6547*** (5.40)	−0.0544* (−1.89)	−0.0546* (−1.90)
$D \times \Delta \ln \text{Rev}$	−5.3501*** (−3.58)	0.0994 (0.79)	0.1013 (0.81)
$\text{ESG} \times D \times \Delta \ln \text{Rev}$	−0.0003 (−0.24)	0.0030 (0.69)	−0.0018 (−1.52)
Constant	0.1037 (0.76)	0.2056** (2.53)	0.2068** (2.56)
Controls	Yes	Yes	Yes
Year	Yes	Yes	Yes
Industry	Yes	Yes	Yes
Observations	1399	1399	1399
Adjusted R ²	0.379	0.025	0.023

Notes: This table reports the results of the placebo test. Column (1) presents the regression results by randomly matching customers to suppliers. Column (2) shows the regression results by randomly matching suppliers to customers. Column (3) shows the regression results by simultaneously randomly matching both customers and suppliers. Below the coefficient estimates, in parentheses, are t-values. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

We employ the net optimism score (OPSC^2) to measure management's optimistic expectations based on the directors' statements in the annual report (Krisnadewi et al., 2023). The larger the OPSC, the stronger the positive inclination of sentiment. In terms of agency problems, referring to Yermack (2006), we use executive perquisite consumption to measure agency costs (Perk). The higher the executive perquisite consumption, the higher the management's opportunistic behavior. We measure adjustment costs using capital intensity (AI) (Banker et al., 2014).

We group the data based on the median of OPSC, Perk, and AI and perform regressions using model (1). The regression results are shown in Table 4. In Column (1), the coefficient of $\text{ESG} \times D \times \Delta \ln \text{Rev}$ is significantly positive at the 1 % level. In Column (2), the coefficient of $\text{ESG} \times D \times \Delta \ln \text{Rev}$ is not significant. This suggests that ESG disclosures by customer firms can reduce suppliers' cost stickiness by lowering optimistic expectations on the part of the suppliers' management. In Columns (3)–(6), coefficients of $\text{ESG} \times D \times \Delta \ln \text{Rev}$ are not significant, indicating that ESG disclosures by customer firms cannot reduce suppliers' cost stickiness by lowering agency problems or adjustment costs.

When management holds optimistic expectations for the future, they often make greater investments in working capital, such as holding more inventory. We measure the management's optimistic expectations (OPT_IN) using the ratio of the firm's inventory turnover to the median value of the industry. Higher OPT_IN values indicate less aggressive investments by management, that is, lower optimistic expectations for the future. We conduct a regression on ESG and OPT_IN, and the results, as shown in column (1) of Table 5, indicating the coefficient of ESG is significantly positive at the 1 % level, suggesting that ESG disclosures by customer firms can lower the optimistic expectations of the suppliers' management.

Furthermore, we categorize the information revealed under ESG disclosure as whether they disclose creditor protection (CRE), environmental protection (ENV), supplier protection (SUP), work safety (SAFE), shareholder protection (SHOLD), and deficiency (DEF). We conduct regressions on above six variables and OPT_IN, and the results are shown in column (2)–(7) of Table 5. We find that the coefficients for CRE and ENV are significantly positive at the 1 % and 5 % levels respectively, indicating that the information on creditor protection and environmental protection can significantly reduce management's optimistic

² $\text{OPSC} = (\text{Number of positive words} - \text{Number of negative words}) / \text{Total number of words}$

Table 4
Mechanism analysis.

Dependent variable: $\Delta \ln SGA$	Optimistic expectations		Opportunistic behavior		Adjustment cost	
	High (1)	Low (2)	High (3)	Low (4)	Low (5)	High (6)
$\Delta \ln Rev$	0.542*** (7.59)	0.771*** (4.99)	0.603*** (8.57)	0.719*** (3.59)	0.803*** (5.63)	0.407*** (4.03)
$D \times \Delta \ln Rev$	-9.074*** (-3.77)	0.902 (0.82)	-0.251 (-0.08)	-5.818*** (-3.00)	0.556 (0.20)	-8.159*** (-3.22)
$ESG \times D \times \Delta \ln Rev$	0.033** (2.04)	0.011 (1.42)	0.040** (2.06)	0.016** (2.10)	0.022** (2.20)	0.052** (1.98)
Constant	0.292 (0.96)	-0.058 (-0.45)	0.292 (0.96)	-0.058 (-0.45)	0.292 (0.96)	-0.058 (-0.45)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes	Yes
Observations	776	623	741	658	704	695
Adjusted R ²	0.408	0.564	0.320	0.518	0.423	0.468

Notes: This table reports the results of mechanism analysis. Columns (1) and (2) present the regression results according to the management's optimistic expectations. Columns (3) and (4) present the regression results according to the management's opportunistic behavior. Columns (5) and (6) present the regression results according to the adjustment costs. Below the coefficient estimates, in parentheses, are t-values. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 5
Mechanism analysis-Working capital.

Dependent variable:	OPT_IN (1)	OPT_IN (2)	OPT_IN (3)	OPT_IN (4)	OPT_IN (5)	OPT_IN (6)	OPT_IN (7)
ESG	0.152*** (4.57)						
CRE		1.573*** (3.19)					
ENV			1.657** (2.54)				
SUP				-0.012 (-0.02)			
SAFE					-0.449 (-0.87)		
SHOLD						-0.618 (-1.23)	
DEF							-0.224 (-0.18)
Constant	-17.106*** (-3.10)	-13.636** (-2.50)	-12.366** (-2.35)	-14.805*** (-2.65)	-15.706*** (-2.89)	-12.657** (-2.27)	-14.268** (-2.42)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1371	1371	1371	1371	1371	1371	1371
Adjusted R ²	0.183	0.172	0.170	0.167	0.167	0.167	0.167

Notes: This table reports the results of mechanism analysis-working capital. Columns (1) shows the result of the regression on ESG and OPT_IN. Columns (2) shows the result of the regression on CRE and OPT_IN. Columns (3) shows the result of the regression on ENV and OPT_IN. Columns (4) shows the result of the regression on SUP and OPT_IN. Columns (5) shows the result of the regression on SAFE and OPT_IN. Columns (6) shows the result of the regression on SHOLD and OPT_IN. Columns (7) shows the result of the regression on DEF and OPT_IN. Below the coefficient estimates, in parentheses, are t-values. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

expectations.

4.4. Further discussion

In this section, we provide further discussion to investigate whether the relationship between ESG disclosures by customer firms and suppliers' cost stickiness is affected by information asymmetries between customers and suppliers.

We use three proxies to empirically measure the information asymmetry between customers and suppliers. First, we measure the geographic distance for each customer-supplier pair. Suppliers that are more geographically distant from their customers face higher customer information acquisition costs and higher information asymmetry. Second, we measure the social distance by considering whether customers and suppliers are in the same dialect area. It is easier to achieve informal socialization and information exchange in the same dialect area. Third,

we identify the industry distance by examining whether customers and suppliers operate within the same industry. Customer-supplier pairs across industries face higher information asymmetry compared to pairs in the same industry (Darendeli et al., 2022).

Table 6 reports the regression results using geographic distance, social distance, and industry distance as grouping variables. If ESG disclosures by customer firms affect suppliers' cost stickiness by reducing information asymmetry in the supply-chain, the coefficient of $ESG \times D \times \Delta \ln Rev$ should be more pronounced in the group with high information asymmetry. In line with our predictions, the findings in Table 6 show significantly stronger effects (both in terms of economic magnitude and statistical significance) for the far geographic distance group (Columns 1), the greater social distance group (Columns 3), and the different industries group (Columns 5).

Table 6

Regression result of further discussion.

Dependent variable: $\Delta \ln SGA$	Geographic distance		Social distance		Industry distance	
	Far (1)	Near (2)	Far (3)	Near (4)	Far (5)	Near (6)
$\Delta \ln Rev$	0.996*** (6.37)	0.428*** (5.40)	0.507*** (7.36)	0.878*** (4.65)	0.677*** (5.14)	0.418*** (4.81)
$D \times \Delta \ln Rev$	-12.833** (-2.39)	-10.065*** (-4.28)	-1.943 (-0.91)	-5.830** (-2.20)	-1.711 (-0.88)	-6.211* (-1.77)
$ESG \times D \times \Delta \ln Rev$	-0.094*** (-3.48)	0.011 (0.91)	0.030*** (3.03)	-0.025 (-1.46)	0.032*** (3.17)	0.017 (0.58)
Constant	0.018 (0.23)	-0.091 (-1.10)	0.172 (1.13)	-1.537* (-1.91)	0.100 (0.72)	0.044 (0.27)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes	Yes
Observations	762	194	1143	256	1104	295
Adjusted R ²	0.3390	0.8393	0.3386	0.6631	0.4453	0.3545

Notes: This table reports the results for the moderating effect of information asymmetry between customers and suppliers. Columns (1) and (2) present the regression results according to the geographic distance. Columns (3) and (4) present the regression results according to the social distance. Columns (5) and (6) present the regression results according to the industry distance. Below the coefficient estimates, in parentheses, are t-values. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

5. Conclusion

We examine the effect of ESG disclosures by customer firms on suppliers' cost stickiness in a supply-chain setting. Our empirical analysis suggests that suppliers' cost stickiness decreases with ESG disclosure by their customer firms. Through mechanism analyses, we attribute this inverse relationship to lowered optimistic expectations on the part of the suppliers' management as the underlying mechanism. Furthermore, we document that the negative relationship is stronger when suppliers are more geographically distant from their customers, when suppliers face greater social distance from their customers, and when customers and suppliers do not operate within the same industry. Overall, the evidence supports the notion that ESG disclosures by customer firms provides valuable information to their business partners in facilitating suppliers' cost practices.

Declaration of competing interest

NA

Data availability

Data will be made available on request.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.econlet.2024.111697](https://doi.org/10.1016/j.econlet.2024.111697).

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